For Regular Graphs, Galvin's Kernel Method Works Only on Bipartite Graphs

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Abstract

Galvin proved the List Edge Colouring Conjecture for bipartite graphs. His proof gives, for every bipartite graph G of maximum degree Δ, an orientation of the line graph $L(G)$ in which every induced subgraph has a kernel and every vertex has out-degree at most $\Delta - 1$. The kernel lemma of Bondy, Boppana and Siegel then turns this orientation into a list edge colouring of G from any lists of size Δ. We prove the reverse direction for regular graphs: if G is a simple k-regular graph and $L(G)$ has such an orientation with every out-degree at most $k - 1$, then G is bipartite. With Galvin's theorem this is a characterisation. We also prove that on a simple regular bipartite graph every such orientation is one of Galvin's own orientations. So, on simple regular graphs, the kernel method with lists of size Δ is exactly Galvin's theorem and nothing more, and it cannot prove the List Edge Colouring Conjecture for any non-bipartite regular graph. The proof is elementary. It is written in very small steps, and every fact it uses is proved here from the definitions, so that it can be checked line by line.

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1 How to read this document

Purpose. This document exists so that the proof of Theorem A can be checked for correctness. Every step is written out, and every step says, in square brackets and italics, which earlier definition or lemma justifies it. A reader who checks each bracket has checked the proof.

What is proved and what is cited. Two results of other authors are stated and *not* proved here: Galvin's theorem (Cited Theorem 2.6) and the kernel lemma (Cited Theorem 2.4). They are labelled "Cited Theorem". They explain why the main theorem matters and give the "if" half of Corollary B, but **the proof of the main theorem uses neither of them**. Everything the main proof uses is proved in Sections 3 to 5, from the definitions. Section 19 also describes results of other papers; those descriptions are cited, not proved.

Layout.

- Section 2 states the problem and the results, says what the main theorem does not claim, and explains why it is not automatic.
- Sections 3, 4 and 5 prove every background fact the proof needs: small facts about numbers, graphs and directed graphs.
- Section 6 fixes the hypothesis of the main proof.
- Sections 7 to 14 are the proof, in eight steps. Section 15 puts the steps together.
- Section 16 runs the proof on small graphs.
- Sections 17 and 18 prove two extensions: rigidity, and orientations with arcs in both directions.
- Section 19 explains the impact of the result. Section 20 reports an independent computer check, and Section 21 reports what was checked in the literature.

Conventions. All graphs and digraphs are finite. $\mathbb{Z}$ is the set of integers. For an integer $\ell \geq 1$, "indices modulo ℓ" means that $a_{\ell+1}$ is a_1, a_0 is a_ℓ, and so on. "Iff" means "if and only if". The symbol $\equiv$ means congruence modulo 2 unless stated otherwise. A set with one element is written $\{x\}$, and $|S|$ is the number of elements of a set S.

Relation to the earlier write-up. The repository note `lecc-small/03-regular-bipartite-theorem.md` proves the same theorem in nine steps. Steps 1 to 3 here are the same as there. From Step 4 on, this document follows a shorter route: the orientation O of Section 10 also orients the edges of middle rank, in both directions, so it is balanced for every k, and a single parity argument (Sections 12 to 14) replaces the separate treatment of k even and k odd in Steps 7 to 9 of the note. The two proofs prove the same statement.

2 The problem and the results

2.1 List edge colouring

Let G be a graph with vertex set V and edge set E (the precise definitions are in Section 4). Two edges are *adjacent* if they share an end.

Definition 2.1 (edge colourings)**.** (a) A *proper edge colouring* of G is a map c from E to some set of colours such that adjacent edges get different colours. The *chromatic index* $\chi'(G)$ is the least number of colours in a proper edge colouring.

(b) A *list assignment* gives every edge e a finite set $C(e)$ of colours, its *list*. A proper edge colouring c *respects* the lists if $c(e) \in C(e)$ for every edge e.

(c) The *list chromatic index* $\mathrm{ch}'(G)$ is the least integer t such that for *every* list assignment with $|C(e)| \geq t$ for all e there is a proper edge colouring respecting the lists.

If all lists are equal to the same set of t colours, a colouring respecting the lists is just a proper edge colouring with t colours. So $\chi'(G) \leq \mathrm{ch}'(G)$ for every graph.

Remark 2.2 (context, cited). Edges at one vertex need different colours, so $\chi'(G) \geq \Delta(G)$, the maximum degree. Vizing's theorem [14] says $\chi'(G) \leq \Delta(G) + 1$ for simple graphs. A graph is of *class one* if $\chi'(G) = \Delta(G)$ and of *class two* if $\chi'(G) = \Delta(G) + 1$. König's theorem [9] says every bipartite graph is of class one. Nothing in the proof of the main theorem uses these facts.

List Edge Colouring Conjecture (LECC). For every graph G, $\mathrm{ch}'(G) = \chi'(G)$.

The conjecture is open. It is known for bipartite graphs (Galvin [6]), for d-regular d-edge-colourable planar graphs (Ellingham and Goddyn [5]), for complete graphs K_{p+1} with p an odd prime (Schauz [12]), and for several other families.

2.2 Kernels and the kernel lemma

The definitions of digraph, orientation, induced subdigraph and out-degree are in Section 5.

Definition 2.3 (kernel)**.** Let D be a digraph with vertex set X. A *kernel* of D is a set $K \subseteq X$ such that

(K1) *(independent)* no arc of D has both ends in K, and

(K2) *(absorbing)* every vertex $x \in X \setminus K$ has an out-neighbour in K, that is, there is $y \in K$ with an arc $x \to y$.

D is *kernel-perfect* if for every nonempty $S \subseteq X$ the induced subdigraph $D[S]$ has a kernel.

This is the definition used in Proofs from THE BOOK [1, Chapter 33, Definition 1].

Cited Theorem 2.4 (kernel lemma; Bondy, Boppana and Siegel; see [1, Chapter 33, Lemma 1])**.** *Let D be a kernel-perfect digraph with vertex set X, and let every $x \in X$ have a list $C(x)$ with $|C(x)| \geq d^+(x) + 1$. Then the vertices can be coloured from their lists so that the two ends of every arc get different colours.*

A proper edge colouring of G is the same thing as a proper vertex colouring of the line graph $L(G)$ (Definition 4.10): the vertices of $L(G)$ are the edges of G, adjacent when they share an end. This gives the method.

Definition 2.5 (certificate)**.** Let G be a graph and $k \geq 1$ an integer. A *k-certificate* for G is an orientation D of $L(G)$ such that

(C1) D is kernel-perfect, and

(C2) $d^+_D(e) \leq k - 1$ for every edge e of G.

By Cited Theorem 2.4, applied to D with lists of size $k \geq d^+(e) + 1$, a k-certificate proves $\mathrm{ch}'(G) \leq k$. When $k = \chi'(G)$ this proves the LECC for G. We call this route *the kernel method with lists of size k*. It is the method of Galvin [6].

2.3 Galvin's theorem: the forward direction

Cited Theorem 2.6 (Galvin [6]). *Let G be a bipartite multigraph with maximum degree Δ. Then $L(G)$ has a kernel-perfect orientation with every out-degree at most $\Delta - 1$. In particular G has a Δ-certificate, and $\mathrm{ch}'(G) = \chi'(G) = \Delta$.*

The orientation is built from a proper edge colouring c with colours $0, 1, \ldots, \Delta - 1$ and a bipartition (A, B): at a vertex of A every edge points to the edges of smaller colour, and at a vertex of B to the edges of larger colour (Definition 17.1). Kernel-perfectness comes from the Gale and Shapley stable matching theorem. An exposition for the $n \times n$ array is [1, Chapter 33]; the same argument works for every bipartite graph. We do not reprove Galvin's theorem.

2.4 The results of this document

Theorem A (the reverse direction). *Let $k \geq 1$ and let G be a simple k-regular graph. If G has a k-certificate, then G is bipartite.*

Corollary B (characterisation). *Let $k \geq 1$ and let G be a simple k-regular graph. Then $L(G)$ has a kernel-perfect orientation with every out-degree at most $k - 1$ if and only if G is bipartite.*

"If" is Galvin's theorem; "only if" is Theorem A. The proof is in Section 15.

Theorem C (rigidity). *Let G be a simple k-regular bipartite graph. Every k-certificate for G is a Galvin orientation D_c for exactly one proper edge colouring c with colours $0, \ldots, k - 1$.*

This is Theorem 17.3 in Section 17.

Proposition D (arcs in both directions). *Theorem A stays true if the certificate is allowed to join adjacent edges by arcs in both directions.*

This is Proposition 18.1 in Section 18.

2.5 What the main theorem does not say

Theorem A is about the *method*, not about the *number* $\mathrm{ch}'(G)$.

- It does **not** say that $\mathrm{ch}'(G) > \Delta$ for non-bipartite regular G. That stronger statement is false. K_4 is 3-regular and contains a triangle, so it is not bipartite, yet $\mathrm{ch}'(K_4) = 3$: this is Schauz's theorem for K_{p+1} with $p = 3$ [12], and also the theorem of Ellingham and Goddyn, since K_4 is planar and has a proper edge colouring with 3 colours [5]. Theorem A says only that no 3-certificate for K_4 exists, so the kernel method cannot be the proof of $\mathrm{ch}'(K_4) = 3$.
- It says nothing new about graphs of class two. If $\chi'(G) = k + 1$, then $\mathrm{ch}'(G) \geq \chi'(G) = k + 1 > k$, so no proof of $\mathrm{ch}'(G) = k$ exists by any method. The theorem matters for non-bipartite regular graphs of *class one*, where the LECC predicts $\mathrm{ch}'(G) = k$.
- It is about lists of size exactly $\Delta = k$. With lists of size $k + 1$ the kernel method does succeed on some non-bipartite regular graphs (Section 20: K_4 and the triangular prism).
- It is about simple graphs. The multigraph case is open (Remark 7.4).

2.6 Why the reverse direction is not automatic

Galvin's construction needs a bipartition: it points edges one way at vertices of A and the other way at vertices of B. So one might think the reverse direction is obvious, "Galvin's method needs bipartite graphs". It is not obvious, for three reasons.

(i) The kernel lemma (Cited Theorem 2.4) works on *every* digraph. Galvin's recipe is one way to build a certificate. It does not show that no other recipe exists. A non-bipartite graph could, a priori, have a certificate built some other way.

(ii) Certificates on non-bipartite graphs do exist once a hypothesis is dropped. The diamond (K_4 minus an edge) is not bipartite, has maximum degree 3, and has 3-certificates (Example 16.4); it is not regular. K_4 is regular and not bipartite, and has 4-certificates (Section 20); there the lists have size $\Delta + 1$. McDonald [10] gives further examples of both kinds: every "bipartite plus an edge" graph [10, Theorem 3], and the Petersen graph with lists of size $4 = \Delta + 1$ [10, Theorem 4].

(iii) The known negative results are weaker. McDonald [10, Theorem 5] shows that K_n ($n \geq 4$) has no certificate of Galvin's special form; Theorem A excludes *every* orientation, for *every* non-bipartite regular graph.

So the question "can some orientation, not Galvin's, prove $\mathrm{ch}'(G) = \Delta$ for a non-bipartite regular graph by the kernel lemma?" was open as far as we could find, and Theorem A answers it: no. Theorem C adds that on regular bipartite graphs there is no orientation beyond Galvin's either.

2.7 The proof in one paragraph, and the roadmap

Take a k-certificate D. At each vertex u the k edges at u are pairwise adjacent in $L(G)$, so D orders them; this gives each edge a *rank* $0, 1, \ldots, k-1$ at each of its two ends (Step 1). Counting arcs shows every out-degree is exactly $k-1$, which forces the two ranks of every edge to add up to $k-1$ (Step 2). An odd cycle of G along which every vertex ranks its next edge better than its previous edge would become a directed odd cycle in D, which has no kernel; so there is no such "odd ring" (Step 3). Now orient G itself: $u \to v$ when u ranks uv in the better half (Step 4). Every directed cycle of length at least 3 of this orientation is a ring (Step 5), so it has no directed cycle of odd length and no directed closed walk of odd length (Step 6). Each vertex has as many out-arcs as in-arcs, so every arc can be walked back, and the way back always has odd length (Step 7). Then every cycle of G can be turned into a directed closed walk of the same parity, which must be even (Step 8). A graph with no odd cycle is bipartite.

Step	Statement	Uses
1	ranks exist; $d^+(uv) = r_u(v) + r_v(u)$	Lemmas 5.8, 4.11, 5.6
2	$r_u(v) + r_v(u) = k - 1$	Step 1, Lemmas 5.3, 4.11
3	no odd ring	Step 1, Lemmas 4.7, 5.7
4	the orientation O of G is balanced	Step 2, Lemma 3.2
5	directed cycles of O of length ≥ 3 are rings	Steps 1, 2, 4
6	O has no odd directed closed walk	Steps 3, 5, Lemma 5.4
7	every arc of O can be walked back, in an odd number of steps	Steps 4, 6, Lemma 5.5
8	every cycle of G is even	Steps 4, 7, Lemma 3.4
end	G is bipartite	Step 8, Lemma 4.9

3 Preliminaries on numbers

Lemma 3.1 (pairs)**.** *A set with t elements has exactly $t(t-1)/2$ subsets with two elements.*

Proof. (1) The number of ordered pairs (x, y) of distinct elements is $t(t-1)$: there are t choices for x and then $t-1$ choices for $y \neq x$.

(2) Each two-element subset $\{x, y\}$ arises from exactly two ordered pairs, (x, y) and (y, x).

(3) So the number of two-element subsets is $t(t-1)/2$. □

Lemma 3.2 (upper and lower halves)**.** *Let $k \geq 1$ and let $R = \{0, 1, \ldots, k-1\}$. Put*

$$R_{\leq} = \{r \in R : r \leq \tfrac{k-1}{2}\}, \qquad R_{\geq} = \{r \in R : r \geq \tfrac{k-1}{2}\}.$$

Then $|R_{\leq}| = |R_{\geq}|$.

Proof. (1) Define $\phi(r) = k - 1 - r$. If $0 \leq r \leq k-1$ then $0 \leq k-1-r \leq k-1$. So ϕ maps R to R.

(2) $\phi(\phi(r)) = k - 1 - (k - 1 - r) = r$. So ϕ is its own inverse, hence a bijection of R.

(3) For $r \in R$: $r \leq \frac{k-1}{2}$ iff $-r \geq -\frac{k-1}{2}$ iff $k - 1 - r \geq k - 1 - \frac{k-1}{2} = \frac{k-1}{2}$. So $r \in R_{\leq}$ iff $\phi(r) \in R_{\geq}$.

(4) By (2) and (3), ϕ restricts to a bijection from $R_{\leq}$ to $R_{\geq}$. So the two sets have the same size. □

Remark 3.3. In fact $|R_{\leq}| = |R_{\geq}| = \lceil k/2 \rceil$, but only the equality of the two sizes is used. If k is odd, then $\frac{k-1}{2}$ is an integer and lies in both sets. If k is even, the two sets are disjoint.

Lemma 3.4 (parity of a sum of odd numbers)**.** *If $q_1, \ldots, q_\ell$ are odd integers, then $q_1 + \cdots + q_\ell \equiv \ell \pmod 2$.*

Proof. (1) Each q_i is odd, so $q_i = 2s_i + 1$ for some integer s_i.

(2) Then $q_1 + \cdots + q_\ell = 2(s_1 + \cdots + s_\ell) + \ell$, which differs from ℓ by an even number. □

4 Preliminaries on graphs

4.1 Definitions

Definition 4.1 (graph)**.** A *simple graph* G is a pair (V, E) of a finite set V of *vertices* and a set E of two-element subsets of V, the *edges*. The edge $\{u, v\}$ is written uv (so $uv = vu$), and u, v are its *ends*. Because E is a *set* of *two-element* subsets:

- there are no loops (an edge has two different ends), and
- there are no parallel edges (two vertices are joined by at most one edge).

A *multigraph* may have parallel edges; it is not a simple graph.

Definition 4.2 (neighbours, stars, degree, regular)**.** Let $G = (V, E)$ be simple and $u \in V$.

- v is a *neighbour* of u if $uv \in E$. $N(u)$ is the set of neighbours.

- The *star* of u is $E(u) = \{e \in E : u \in e\}$, the set of edges at u.
- The *degree* of u is $d(u) = |E(u)|$. The map $v \mapsto uv$ is a bijection from $N(u)$ to $E(u)$, so $d(u) = |N(u)|$ as well.
- G is *k-regular* if $d(u) = k$ for every $u \in V$. $\Delta(G)$ is the maximum degree.

Definition 4.3 (walks and cycles)**.** Let $G = (V, E)$ be simple.

- A *walk* of *length* $L \geq 0$ from w_0 to w_L is a sequence $(w_0, w_1, \ldots, w_L)$ of vertices with $w_j w_{j+1} \in E$ for $0 \leq j < L$. Vertices may repeat. It is *closed* if $w_L = w_0$.
- A *cycle* of *length* ℓ is a sequence $(a_1, \ldots, a_\ell)$ of $\ell \geq 3$ *distinct* vertices such that $a_i a_{i+1} \in E$ for every i, indices modulo ℓ. Its *edges* are $e_i = a_i a_{i+1}$ for $i = 1, \ldots, \ell$. It is *odd* or *even* according to ℓ.

Definition 4.4 (bipartite)**.** $G = (V, E)$ is *bipartite* if V can be split into two disjoint sets A and B (with $A \cup B = V$) such that every edge has one end in A and one end in B. The pair (A, B) is a *bipartition*.

4.2 Basic lemmas

Lemma 4.5 (two edges share at most one end)**.** *In a simple graph, two distinct edges have at most one common end.*

Proof. (1) Let $e \neq f$ be edges and suppose they have two common ends $u \neq v$.

(2) e has exactly two elements *[Definition 4.1]* and contains u and v, so $e = \{u, v\}$. In the same way $f = \{u, v\}$.

(3) So $e = f$, contradicting $e \neq f$. □

This is the only place where the absence of parallel edges enters. Everything later that needs simplicity goes through this lemma.

Lemma 4.6 (handshake)**.** *For a simple graph, $\sum_{u \in V} d(u) = 2|E|$. If G is k-regular with n vertices, then $nk = 2|E|$.*

Proof. (1) Count the pairs (u, e) with $u \in V$, $e \in E$ and $u \in e$.

(2) Grouping by u: vertex u is in $|E(u)| = d(u)$ such pairs. Total $\sum_u d(u)$.

(3) Grouping by e: each edge has exactly two ends, so it is in exactly two pairs. Total $2|E|$.

(4) The two totals count the same set, so they are equal. If $d(u) = k$ for all n vertices, the left side is nk. □

Lemma 4.7 (edges of a cycle)**.** *Let $(a_1, \ldots, a_\ell)$ be a cycle with edges $e_i = a_i a_{i+1}$ (indices modulo ℓ). Then:*

(a) the edges $e_1, \ldots, e_\ell$ are pairwise distinct;

(b) for $i \neq j$, the edges e_i and e_j have a common end iff $j \equiv i + 1$ or $j \equiv i - 1 \pmod{\ell}$;

(c) the only common end of e_{i-1} and e_i is a_i.

Proof. Throughout, recall that $a_1, \ldots, a_\ell$ are distinct and $\ell \geq 3$, so two vertices a_p, a_q are equal iff $p \equiv q \pmod{\ell}$.

(a)

(1) Suppose $e_i = e_j$ with $i \not\equiv j$.

(2) $a_i \in e_i = e_j = \{a_j, a_{j+1}\}$, so $a_i = a_j$ or $a_i = a_{j+1}$. The first means $i \equiv j$, which is excluded. So $a_i = a_{j+1}$, that is, $j \equiv i - 1$.

(3) $a_{i+1} \in e_j$, so $a_{i+1} = a_j$ or $a_{i+1} = a_{j+1}$. The second means $i \equiv j$, which is excluded. So $a_{i+1} = a_j$, that is, $j \equiv i + 1$.

(4) By (2) and (3), $i - 1 \equiv i + 1 \pmod{\ell}$, so ℓ divides 2. This contradicts $\ell \geq 3$.

(b)

(1) $e_i = \{a_i, a_{i+1}\}$ and $e_j = \{a_j, a_{j+1}\}$. A common end is a vertex in both, so one of these holds: $a_i = a_j$, $a_i = a_{j+1}$, $a_{i+1} = a_j$, $a_{i+1} = a_{j+1}$.

(2) These mean, respectively, $i \equiv j$, $i \equiv j + 1$, $i + 1 \equiv j$, $i + 1 \equiv j + 1$ (modulo ℓ).

(3) The first and the last say $i \equiv j$, which is excluded. The second says $j \equiv i - 1$. The third says $j \equiv i + 1$.

(4) Conversely, if $j \equiv i + 1$ then $a_{i+1} = a_j$ is a common end, and if $j \equiv i - 1$ then $a_i = a_{j+1}$ is a common end.

(c)

(1) a_i is in $e_{i-1} = \{a_{i-1}, a_i\}$ and in $e_i = \{a_i, a_{i+1}\}$.

(2) By (a), $e_{i-1} \neq e_i$. By Lemma 4.5 they have at most one common end. So a_i is the only one. □

Lemma 4.8 (odd closed walks contain odd cycles)**.** *If a simple graph has a closed walk of odd length, then it has an odd cycle.*

Proof. By strong induction on the length L. Let $W = (w_0, \ldots, w_L)$ be a closed walk with L odd, and assume the lemma for all odd lengths smaller than L.

(1) $L \neq 1$: a closed walk of length 1 would need $w_0 w_1 = w_0 w_0 \in E$, a loop, and there are no loops *[Definition 4.1]*. So $L \geq 3$, as L is odd.

(2) *Case 1: $w_0, \ldots, w_{L-1}$ are distinct.* Then $(w_0, \ldots, w_{L-1})$ is a sequence of $L \geq 3$ distinct vertices, consecutive ones are joined by edges, and so are w_{L-1} and $w_L = w_0$. This is a cycle of odd length L *[Definition 4.3]*.

(3) *Case 2: $w_i = w_j$ for some $0 \leq i < j \leq L - 1$.* Split W into

$$W_1 = (w_i, w_{i+1}, \ldots, w_j), \qquad W_2 = (w_0, \ldots, w_i, w_{j+1}, \ldots, w_L).$$

W_1 is a closed walk (it starts at w_i and ends at $w_j = w_i$) of length $j - i$. W_2 is a walk, because $w_i w_{j+1} = w_j w_{j+1} \in E$, and it is closed, of length $i + (L - j)$.

(4) Both lengths are between 1 and $L - 1$, and they add up to L, which is odd. So one of them is odd and smaller than L.

(5) By the induction hypothesis, that shorter closed walk gives an odd cycle. □

Lemma 4.9 (no odd cycle implies bipartite)**.** *If a simple graph G has no odd cycle, then G is bipartite.*

Proof. (1) Call x and y *connected* if there is a walk from x to y. This is an equivalence relation: (x) is a walk of length 0; a walk reversed is a walk; two walks $x \to y$ and $y \to z$ joined at y give a walk $x \to z$. Its classes are the *components*.

(2) In each component choose one vertex, its *root*. For each vertex x, let $s(x)$ be the root of its component and $\delta(x)$ the least length of a walk from $s(x)$ to x. It exists, since such walks exist.

(3) Put $A = \{x : \delta(x) \text{ even}\}$ and $B = \{x : \delta(x) \text{ odd}\}$. They are disjoint and cover V.

(4) Let $xy \in E$; we show x, y are not both in A and not both in B. Suppose $\delta(x) \equiv \delta(y)$. The edge xy is a walk, so x, y are in one component and $s(x) = s(y) = s$.

(5) Take a shortest walk P from s to x (length $\delta(x)$) and a shortest walk Q from s to y (length $\delta(y)$). Follow P from s to x, then the edge xy, then Q backwards from y to s. This is a closed walk of length $\delta(x) + 1 + \delta(y)$.

(6) Since $\delta(x) \equiv \delta(y)$, the length is $\equiv 2\delta(x) + 1 \equiv 1$: it is odd.

(7) By Lemma 4.8, G has an odd cycle, contradicting the hypothesis. So every edge has one end in A and one in B, and (A, B) is a bipartition. □

4.3 The line graph

Definition 4.10 (line graph)**.** The *line graph* $L(G)$ of a simple graph $G = (V, E)$ has vertex set E, and two distinct edges $e, f \in E$ are adjacent in $L(G)$ iff they have a common end in G.

To avoid confusion we keep saying *edge* for an element of E, also when it is regarded as a vertex of $L(G)$.

Lemma 4.11 (structure of the line graph)**.** *Let $G = (V, E)$ be simple.*

(a) For each $u \in V$, any two distinct edges of $E(u)$ are adjacent in $L(G)$.

(b) If e, f are adjacent in $L(G)$, they have exactly one common end. Call it $w(e, f)$.

(c) For an edge $e = uv$, the neighbours of e in $L(G)$ are the edges of $(E(u) \setminus \{e\}) \cup (E(v) \setminus \{e\})$, and these two sets are disjoint.

(d) If G is k-regular with n vertices, then $L(G)$ has exactly $nk(k-1)/2 = |E|(k-1)$ edges.

Proof. (a) Two distinct edges of $E(u)$ both contain u, so they have the common end u.

(b) They have at least one common end by Definition 4.10, and at most one by Lemma 4.5.

(c)

(1) An edge $f \neq e$ is a neighbour of e iff f contains u or v, that is, iff $f \in E(u) \setminus \{e\}$ or $f \in E(v) \setminus \{e\}$.

(2) If f were in both sets, f and e would have the two common ends u, v, which Lemma 4.5 forbids. So the sets are disjoint.

(d)

(1) Send every edge $\{e, f\}$ of $L(G)$ to the vertex $w(e, f)$ of G. By (b) this is well defined.

(2) The edges of $L(G)$ sent to a fixed vertex w are exactly the two-element subsets of $E(w)$. Indeed, such an edge $\{e, f\}$ has $e, f \in E(w)$; conversely, any two distinct $e, f \in E(w)$ are adjacent by (a), with common end w, and by (b) $w(e, f) = w$.

(3) $|E(w)| = k$, so by Lemma 3.1 there are $k(k-1)/2$ such subsets for each w.

(4) Summing over the n vertices gives $nk(k-1)/2$ edges of $L(G)$.

(5) By Lemma 4.6, $nk = 2|E|$, so $nk(k-1)/2 = |E|(k-1)$. □

5 Preliminaries on digraphs and kernels

5.1 Definitions

Definition 5.1 (digraph)**.** A *digraph* D is a pair (X, A) of a finite set X and a set A of ordered pairs (x, y) of *distinct* elements of X, the *arcs*. The arc (x, y) is written $x \to y$. So there are no loops and no two equal arcs, but $x \to y$ and $y \to x$ may both be arcs.

- y is an *out-neighbour* of x if $x \to y$. The *out-degree* $d_D^+(x)$ is the number of out-neighbours, and the *in-degree* $d_D^-(x)$ the number of z with $z \to x$.
- D is *balanced* if $d_D^+(x) = d_D^-(x)$ for every $x \in X$.
- For $S \subseteq X$, the *induced subdigraph* $D[S]$ has vertex set S and as arcs all arcs of D with both ends in S.
- An *orientation* of a graph H with vertex set X is a digraph on X such that for every edge xy of H exactly one of $x \to y$, $y \to x$ is an arc, and there are no other arcs.
- A *tournament* is an orientation of a complete graph: between any two distinct vertices there is exactly one arc.

Definition 5.2 (directed walks and cycles)**.** Let $D = (X, A)$ be a digraph.

- A *directed walk* of *length* $L \geq 0$ from w_0 to w_L is a sequence $(w_0, \ldots, w_L)$ with $w_j \to w_{j+1}$ for $0 \leq j < L$. It is *closed* if $w_L = w_0$.
- If W is a directed walk from x to y and W' one from y to z, their *concatenation* WW' (follow W, then W') is a directed walk from x to z, and its length is the sum of the two lengths.
- A *directed cycle* of length $\ell \geq 2$ is a sequence $(x_1, \ldots, x_\ell)$ of distinct vertices with $x_i \to x_{i+1}$ for every i, indices modulo ℓ.

5.2 Counting and walking

Lemma 5.3 (out-degrees count arcs)**.** $\sum_{x \in X} d_D^+(x) = |A|$. *If D is an orientation of a graph H, then $|A|$ is the number of edges of H.*

Proof. Every arc $x \to y$ is counted exactly once on the left, in $d_D^+(x)$. If D is an orientation of H, each edge of H gives exactly one arc and every arc comes from an edge. □

Lemma 5.4 (odd closed directed walks contain odd directed cycles)**.** *If a digraph has a closed directed walk of odd length, then it has a directed cycle of odd length. Every directed cycle of odd length has length at least* 3*.*

Proof. The second sentence: a directed cycle has length $\ell \geq 2$ by definition, and an odd $\ell \geq 2$ is at least 3.

The first sentence, by strong induction on the length L of a closed directed walk $W = (w_0, \ldots, w_L)$ with L odd.

(1) $L \neq 1$: that would need an arc $w_0 \to w_0$, and digraphs have no loops *[Definition 5.1]*. So $L \geq 3$.

(2) *Case 1: $w_0, \ldots, w_{L-1}$ are distinct.* Then $(w_0, \ldots, w_{L-1})$ is a directed cycle of odd length L *[Definition 5.2; the last arc is $w_{L-1} \to w_L = w_0$]*.

(3) *Case 2: $w_i = w_j$ for some $0 \leq i < j \leq L-1$.* Then $W_1 = (w_i, \ldots, w_j)$ is a closed directed walk of length $j - i$, and $W_2 = (w_0, \ldots, w_i, w_{j+1}, \ldots, w_L)$ is a closed directed walk of length $L - (j - i)$, since $w_i \to w_{j+1}$ is the arc $w_j \to w_{j+1}$.

(4) Both lengths lie between 1 and $L - 1$ and add up to the odd number L, so one of them is odd and shorter than L. The induction hypothesis applies to it. □

Lemma 5.5 (in a balanced digraph every arc can be walked back)**.** *Let $D = (X, A)$ be a balanced digraph and $x \to y$ an arc. Then there is a directed walk from y to x.*

Proof. (1) Let R be the set of vertices z such that some directed walk goes from y to z. Then $y \in R$ (walk of length 0).

(2) *No arc leaves R:* if $z \in R$ and $z \to w$, then $w \in R$, since a walk from y to z followed by the arc $z \to w$ is a walk from y to w.

(3) Let a be the number of arcs with both ends in R, and b the number of arcs with tail outside R and head in R.

(4) $\sum_{z \in R} d_D^+(z)$ counts the arcs with tail in R. By (2) all of them have head in R. So the sum is a.

(5) $\sum_{z \in R} d_D^-(z)$ counts the arcs with head in R. Their tails are in R or not. So the sum is $a + b$.

(6) D is balanced, so the sums in (4) and (5) are equal: $a = a + b$, hence $b = 0$.

(7) The arc $x \to y$ has head $y \in R$. If $x \notin R$, it would be counted in $b = 0$. So $x \in R$: there is a directed walk from y to x. □

5.3 Kernels

Kernels and kernel-perfectness were defined in Definition 2.3.

Lemma 5.6 (kernel-perfectness is inherited)**.** *If D is kernel-perfect and $T \subseteq X$, then $D[T]$ is kernel-perfect.*

Proof. For nonempty $S \subseteq T$, the digraph $(D[T])[S]$ has vertex set S and the arcs of $D[T]$ with both ends in S, which are the arcs of D with both ends in S. So $(D[T])[S] = D[S]$, which has a kernel because D is kernel-perfect. □

Lemma 5.7 (an odd directed cycle has no kernel)**.** *Let $\ell \geq 3$ be odd and let F be a digraph whose vertices are ℓ distinct elements $s_1, \ldots, s_\ell$ and whose arcs are exactly $s_i \to s_{i+1}$ for $i = 1, \ldots, \ell$ (indices modulo ℓ). Then F has no kernel.*

Proof. Suppose K is a kernel of F. Put $\kappa_i = 1$ if $s_i \in K$ and $\kappa_i = 0$ otherwise.

(1) The only out-neighbour of s_i is s_{i+1} *[the list of arcs].*

(2) If $s_i \notin K$, then s_i has an out-neighbour in K *[(K2)]*, so $s_{i+1} \in K$ *[(1)].*

(3) If $s_i \in K$, then $s_{i+1} \notin K$, since the arc $s_i \to s_{i+1}$ would otherwise have both ends in K *[(K1)].*

(4) By (2) and (3), $\kappa_{i+1} = 1 - \kappa_i$, so $\kappa_{i+1} \equiv \kappa_i + 1$ for every i.

(5) Applying (4) ℓ times, starting at $i = 1$: $\kappa_{1+\ell} \equiv \kappa_1 + \ell$.

(6) But $s_{1+\ell} = s_1$ (indices modulo ℓ), so $\kappa_{1+\ell} = \kappa_1$, and (5) gives $\ell \equiv 0$. This contradicts ℓ odd. □

In words: going around the cycle, membership in K must alternate, and an odd cycle cannot be coloured alternately.

Lemma 5.8 (kernel-perfect tournaments are linear orders)**.** *Let T be a tournament with $t \geq 1$ vertices in which every induced subdigraph on a nonempty vertex set has a kernel. Then the vertices of T can be listed as $x_0, x_1, \ldots, x_{t-1}$ so that, for all $i \neq j$,*

$$x_i \to x_j \text{ is an arc} \iff i > j.$$

In particular $d^+_T(x_i) = i$, and the out-degrees of the vertices of T are exactly $0, 1, \ldots, t-1$, each once.

Proof. By induction on t. For $t = 1$ there are no arcs, and x_0 is the only vertex. Let $t \geq 2$.

(1) $T = T[V(T)]$ has a kernel K *[hypothesis].*

(2) $|K| \leq 1$: two distinct vertices of K are joined by an arc, as T is a tournament, which (K1) forbids.

(3) $K \neq \varnothing$: otherwise any vertex (there is one, $t \geq 1$) lies outside K and must have an out-neighbour in $K = \varnothing$ *[(K2)]*, which is impossible.

(4) So $K = \{s\}$ for one vertex s, and by (K2) every other vertex x has the arc $x \to s$. Put $x_0 = s$.

(5) Let $T' = T[V(T) \setminus \{s\}]$. It is a tournament with $t - 1 \geq 1$ vertices, and each of its induced subdigraphs is an induced subdigraph of T (as in Lemma 5.6), so it has a kernel.

(6) By the induction hypothesis, list the vertices of T' as $x_1, \ldots, x_{t-1}$ with $x_i \to x_j$ iff $i > j$, for $i, j \geq 1$. Arcs of T' are exactly the arcs of T between these vertices, so this holds in T.

(7) Pairs involving x_0: for $i \geq 1$, $x_i \to x_0$ by (4), which agrees with $i > 0$. And $x_0 \to x_i$ is not an arc, because exactly one of the two arcs is present; this agrees with $0 > i$ being false.

(8) The out-neighbours of x_i are the x_j with $j < i$; there are i of them. □

In words: a kernel of a tournament is a single vertex that every other vertex points to, a "sink". Remove it and repeat. The order of removal is a linear order, and every vertex points exactly to the vertices removed before it.

6 The setting of the proof

Standing Hypothesis 6.1. From now until the end of Section 15:

- $k \geq 1$ is an integer;
- $G = (V, E)$ is a simple k-regular graph with $n = |V|$ vertices;
- D is a k-certificate for G (Definition 2.5): an orientation of $L(G)$ that is kernel-perfect and has $d_D^+(e) \leq k - 1$ for every $e \in E$.

The goal is to prove that G is bipartite. Arcs of D are written $e \to f$ with $e, f \in E$. Later, in Section 10, we define a second digraph O, on V; its arcs are written $u \to v$ with $u, v \in V$. The letters make clear which one is meant: e, f are edges and u, v, w, a, x, y, z are vertices.

7 Step 1: every vertex ranks its edges

Lemma 7.1 (the star lemma)**.** *For every vertex u, the induced subdigraph $D_u = D[E(u)]$ is a tournament on the k edges at u, and its edges can be listed as $f_0, \ldots, f_{k-1}$ so that, for distinct f_i, f_j, $f_i \to f_j$ in D iff $i > j$.*

Proof. (1) $|E(u)| = d(u) = k \geq 1$ *[G is k-regular].*

(2) Any two distinct $e, f \in E(u)$ are adjacent in $L(G)$ *[Lemma 4.11(a)]*. D is an orientation of $L(G)$, so exactly one of $e \to f$, $f \to e$ is an arc of D, and it is an arc of D_u because both ends lie in $E(u)$. So D_u is a tournament.

(3) D_u is kernel-perfect *[Lemma 5.6, since D is]*. So every induced subdigraph of D_u on a nonempty set has a kernel.

(4) Lemma 5.8 applied to D_u with $t = k$ gives the list. □

Definition 7.2 (rank, preference)**.** For a vertex u and an edge $e \in E(u)$, the *rank of e at u* is

$$r_u(e) = d^+_{D_u}(e) = \big|\{f \in E(u) : e \to f\}\big|,$$

the number of edges at u that e points to. For a neighbour v of u we write $r_u(v) = r_u(uv)$. If v, w are neighbours of u with $r_u(v) < r_u(w)$, we say that *u prefers v to w*.

Rank 0 is the best: an edge of rank 0 at u points to no other edge at u, and all other edges at u point to it. Note that every edge uv has *two* ranks, $r_u(v)$ at one end and $r_v(u)$ at the other, and they need not be equal.

Lemma 7.3 (properties of ranks)**.** *(a) For each u, the map $e \mapsto r_u(e)$ is a bijection from $E(u)$ onto $\{0, 1, \ldots, k-1\}$. Equivalently, $v \mapsto r_u(v)$ is a bijection from $N(u)$ onto $\{0, \ldots, k-1\}$.*

(b) For distinct $e, f \in E(u)$: $e \to f$ in D iff $r_u(e) > r_u(f)$.

(c) For every edge $e = uv$: $d_D^+(e) = r_u(v) + r_v(u)$.

Proof. (a)

(1) List $E(u)$ as $f_0, \ldots, f_{k-1}$ as in Lemma 7.1.

(2) f_i points to exactly the f_j with $j < i$, so $r_u(f_i) = i$.

(3) So the ranks of $f_0, \dots, f_{k-1}$ are $0, \dots, k-1$: every value once.

(4) The second form follows because $v \mapsto uv$ is a bijection from $N(u)$ to $E(u)$.

(b) With $e = f_i$ and $f = f_j$: $e \to f$ iff $i > j$ *[Lemma 7.1]* iff $r_u(e) > r_u(f)$ *[(a)(2)]*.

(c)

(1) The out-neighbours of e in D are among its neighbours in $L(G)$, which form the disjoint union $(E(u) \setminus \{e\}) \cup (E(v) \setminus \{e\})$ *[Lemma 4.11(c)]*.

(2) The out-neighbours of e in $E(u) \setminus \{e\}$ are exactly the $f \in E(u)$ with $e \to f$, and there are $r_u(e) = r_u(v)$ of them *[Definition 7.2]*.

(3) In the same way, there are $r_v(e) = r_v(u)$ out-neighbours in $E(v) \setminus \{e\}$.

(4) The two sets are disjoint, so $d_D^+(e) = r_u(v) + r_v(u)$. □

Remark 7.4 (where simplicity is used). Part (c) is where it matters that G has no parallel edges. If e and f were parallel edges between u and v, the single arc between them would be counted both in $r_u(e)$ and in $r_v(e)$, and it would have to agree with the order at u and with the order at v at once. Then (c) fails, and so does Step 2. This is why the theorem is stated for simple graphs.

8 Step 2: the budget is used exactly

Lemma 8.1 (tightness)**.** *Every edge e has $d_D^+(e) = k-1$. Equivalently, for every edge uv,*

$$r_u(v) + r_v(u) = k - 1.$$

Proof. (1) $\sum_{e \in E} d_D^+(e)$ equals the number of edges of $L(G)$ *[Lemma 5.3, as D is an orientation of $L(G)$]*.

(2) That number is $|E|(k-1)$ *[Lemma 4.11(d)]*.

(3) Each of the $|E|$ terms $d_D^+(e)$ is at most $k-1$ *[(C2)]*.

(4) If some term were strictly less than $k-1$, the sum would be strictly less than $|E|(k-1)$, contradicting (1) and (2). So every term equals $k-1$.

(5) For $e = uv$, $d_D^+(e) = r_u(v) + r_v(u)$ *[Lemma 7.3(c)]*. So $r_u(v) + r_v(u) = k-1$. □

In words: the two ends of an edge always "disagree in a balanced way". If u gives uv a good rank, v must give it a correspondingly bad one.

Remark 8.2 (where regularity is used, and what fails without it). This step is the only place where regularity is used. For a simple graph G of maximum degree Δ that is not regular, the same count gives

$$\sum_e d_D^+(e) = \sum_{w \in V} \frac{d(w)(d(w)-1)}{2}, \qquad \sum_e (\Delta - 1) = |E|(\Delta - 1),$$

and the difference, the *slack*, equals

$$\sum_{uv \in E} \Big(\Delta - \frac{d(u) + d(v)}{2} \Big).$$

(Write $\frac{d(w)(d(w)-1)}{2} = \sum_{e \in E(w)} \frac{d(w)-1}{2}$ and regroup by edges.) When the slack is positive, the out-degrees need not all equal $\Delta - 1$, Step 2 fails, and the conclusion fails too: the diamond has slack 2 and does have 3-certificates (Example 16.4).

9 Step 3: there is no odd ring

Definition 9.1 (ring)**.** A *ring* is a cycle $(a_1, \dots, a_\ell)$ of G such that every a_i prefers its successor to its predecessor:

$$r_{a_i}(a_{i+1}) < r_{a_i}(a_{i-1}) \qquad \text{for every } i \text{ (indices modulo } \ell).$$

An *odd ring* is a ring of odd length.

Lemma 9.2 (no odd ring)**.** *G has no odd ring.*

Proof. Suppose $(a_1, \dots, a_\ell)$ is an odd ring. Let $e_i = a_i a_{i+1}$ and $S = \{e_1, \dots, e_\ell\}$.

(1) The edges $e_1, \dots, e_\ell$ are distinct, so $|S| = \ell$ *[Lemma 4.7(a)]*.

(2) Two edges e_i, e_j of S are adjacent in $L(G)$ iff $j \equiv i \pm 1 \pmod{\ell}$ *[Lemma 4.7(b)]*. Since D is an orientation of $L(G)$, $D[S]$ has no arc between e_i and e_j unless $j \equiv i \pm 1$, and exactly one arc between e_{i-1} and e_i.

(3) Fix i. The edges $e_{i-1} = a_{i-1}a_i$ and $e_i = a_i a_{i+1}$ both lie in $E(a_i)$ *[Lemma 4.7(c)]*.

(4) Their ranks at a_i are $r_{a_i}(e_i) = r_{a_i}(a_{i+1})$ and $r_{a_i}(e_{i-1}) = r_{a_i}(a_{i-1})$, and the ring condition says the first is smaller.

(5) So $r_{a_i}(e_{i-1}) > r_{a_i}(e_i)$, and $e_{i-1} \to e_i$ is the arc of D between them *[Lemma 7.3(b)]*.

(6) By (2) and (5), the arcs of $D[S]$ are exactly $e_1 \to e_2 \to \cdots \to e_\ell \to e_1$. (Each unordered pair $\{e_{i-1}, e_i\}$ is met for exactly one i, since $\ell \geq 3$.)

(7) ℓ is odd and at least 3, so $D[S]$ has no kernel *[Lemma 5.7 with $s_i = e_i$]*.

(8) But D is kernel-perfect, so $D[S]$ has a kernel *[(C1)]*. Contradiction. □

Remark 9.3. This is the easy half of the odd-cycle condition of Borodin, Kostochka and Woodall [4], specialised to our situation. It is proved here from the definitions, so nothing is cited.

10 Step 4: an orientation of G itself

Now we leave $L(G)$ and build a digraph on the vertex set V of G.

Definition 10.1 (the digraph O)**.** O is the digraph with vertex set V and an arc $u \to v$ whenever $uv \in E$ and

$$r_u(v) \leq \tfrac{k-1}{2}.$$

In words: u points to v when u ranks the edge uv in its better half.

O has no loops, since $uv \in E$ forces $u \neq v$. So it is a digraph in the sense of Definition 5.1. Some edges may receive arcs in both directions; that is allowed.

Lemma 10.2 (properties of O)**.** *(a) Every arc of O joins the two ends of an edge of G.*

(b) For every edge uv, at least one of $u \to v$ and $v \to u$ is an arc of O.

(c) If $u \to v$ is an arc of O, then $r_v(u) \geq \frac{k-1}{2}$.

(d) O is balanced: $d_O^+(u) = d_O^-(u)$ for every vertex u.

Proof. (a) By Definition 10.1.

(b)

(1) Suppose neither arc is present. Then $r_u(v) > \frac{k-1}{2}$ and $r_v(u) > \frac{k-1}{2}$.

(2) Adding, $r_u(v) + r_v(u) > k - 1$. This contradicts Lemma 8.1.

(c) $r_v(u) = k - 1 - r_u(v)$ *[Lemma 8.1]* $\geq k - 1 - \frac{k-1}{2} = \frac{k-1}{2}$, using $r_u(v) \leq \frac{k-1}{2}$.

(d)

(1) *Out-arcs.* $u \to v$ iff $v \in N(u)$ and $r_u(v) \leq \frac{k-1}{2}$. The map $v \mapsto r_u(v)$ is a bijection from $N(u)$ onto $\{0, \ldots, k-1\}$ *[Lemma 7.3(a)]*. So $d_O^+(u) = |R_\leq|$, in the notation of Lemma 3.2.

(2) *In-arcs.* $v \to u$ iff $v \in N(u)$ and $r_v(u) \leq \frac{k-1}{2}$. By Lemma 8.1, $r_v(u) = k - 1 - r_u(v)$, so the condition reads $k - 1 - r_u(v) \leq \frac{k-1}{2}$, that is, $r_u(v) \geq \frac{k-1}{2}$.

(3) By the same bijection as in (1), $d_O^-(u) = |R_\geq|$.

(4) $|R_\leq| = |R_\geq|$ *[Lemma 3.2]*. So $d_O^+(u) = d_O^-(u)$. □

Remark 10.3. If k is even, $\frac{k-1}{2}$ is not an integer, so no rank equals it, and by Lemma 8.1 exactly one end of each edge has rank below $\frac{k-1}{2}$: O is an orientation of G. If k is odd, an edge gets arcs in both directions exactly when both its ranks equal the middle rank $\frac{k-1}{2}$; each vertex has exactly one edge of middle rank, so these edges form a perfect matching. The proof does not need either fact; Lemma 10.2 is all it uses.

11 Step 5: directed cycles of O are rings

Lemma 11.1. *Let $(a_1, \ldots, a_\ell)$ be a directed cycle of O with $\ell \geq 3$. Then $(a_1, \ldots, a_\ell)$ is a cycle of G, and it is a ring.*

Proof. (1) The a_i are distinct *[Definition 5.2]*, $\ell \geq 3$, and $a_i a_{i+1} \in E$ for every i *[Lemma 10.2(a)]*. So $(a_1, \ldots, a_\ell)$ is a cycle of G *[Definition 4.3]*.

(2) Fix i. The arc $a_i \to a_{i+1}$ gives $r_{a_i}(a_{i+1}) \leq \frac{k-1}{2}$ *[Definition 10.1]*.

(3) The arc $a_{i-1} \to a_i$ gives $r_{a_i}(a_{i-1}) \geq \frac{k-1}{2}$ *[Lemma 10.2(c), with $u = a_{i-1}$ and $v = a_i$]*.

(4) So $r_{a_i}(a_{i+1}) \leq \frac{k-1}{2} \leq r_{a_i}(a_{i-1})$.

(5) $a_{i+1} \neq a_{i-1}$, because the vertices are distinct and $\ell \geq 3$ (so $i + 1 \not\equiv i - 1$). So $r_{a_i}(a_{i+1}) \neq r_{a_i}(a_{i-1})$ *[Lemma 7.3(a): ranks at a_i are distinct]*.

(6) From (4) and (5), $r_{a_i}(a_{i+1}) < r_{a_i}(a_{i-1})$: a_i prefers its successor to its predecessor.

(7) This holds for every i, so the cycle is a ring *[Definition 9.1]*. □

In words: following an arc of O into a vertex means arriving along an edge that the vertex ranks in its worse half, and leaving along an arc means leaving along an edge in its better half. So every vertex on a directed cycle prefers the way out to the way in, which is exactly the ring condition. The only care needed is when both ranks equal the middle value; that cannot happen at one vertex for two different edges.

12 Step 6: O has no odd directed closed walk

Lemma 12.1. *Every closed directed walk of O has even length.*

Proof. (1) Suppose O has a closed directed walk of odd length.

(2) Then O has a directed cycle $(a_1, \ldots, a_\ell)$ of odd length $\ell \geq 3$ *[Lemma 5.4].*

(3) It is a cycle of G and a ring *[Lemma 11.1]*, and ℓ is odd, so it is an odd ring.

(4) This contradicts Lemma 9.2. □

13 Step 7: every arc can be walked back, in an odd number of steps

Lemma 13.1. *Let $x \to y$ be an arc of O. Then there is a directed walk of O from y to x, and every directed walk of O from y to x has odd length.*

Proof. (1) O is balanced *[Lemma 10.2(d)]*, so a directed walk P from y to x exists *[Lemma 5.5].*

(2) Let P be any directed walk from y to x, of length $|P|$. The arc $x \to y$ followed by P is a closed directed walk of length $1 + |P|$ *[Definition 5.2, concatenation].*

(3) This length is even *[Lemma 12.1].* So $|P|$ is odd. □

14 Step 8: every cycle of G is even

Lemma 14.1. *Every cycle of G has even length.*

Proof. Let $(z_1, \ldots, z_\ell)$ be a cycle of G, indices modulo ℓ. For each i we build a directed walk W_i of O from z_i to z_{i+1} of odd length.

(1) $z_i z_{i+1}$ is an edge of G, so $z_i \to z_{i+1}$ or $z_{i+1} \to z_i$ is an arc of O *[Lemma 10.2(b)].*

(2) *If $z_i \to z_{i+1}$ is an arc,* let $W_i = (z_i, z_{i+1})$. Its length is 1, which is odd.

(3) *Otherwise $z_{i+1} \to z_i$ is an arc.* Apply Lemma 13.1 with $x = z_{i+1}$ and $y = z_i$: there is a directed walk W_i from z_i to z_{i+1}, and its length is odd.

(4) Concatenate: $W = W_1 W_2 \cdots W_\ell$ goes from z_1 to z_2, then to z_3, and so on, and finally from z_ℓ to $z_{\ell+1} = z_1$. So W is a closed directed walk of O, of length $|W_1| + \cdots + |W_\ell|$.

(5) Each $|W_i|$ is odd, so $|W| \equiv \ell$ *[Lemma 3.4].*

(6) $|W|$ is even *[Lemma 12.1].* So ℓ is even. □

In words: walk around any cycle of G. Where an edge points the right way in O, use it; where it points the wrong way, take the way back that Step 7 provides. Every piece has odd length, so the closed walk has the same parity as the cycle, and Step 6 says it is even.

15 Proof of the main results

Proof of Theorem A. Let $k \geq 1$, let G be a simple k-regular graph, and let D be a k-certificate for G. Then Standing Hypothesis 6.1 holds, so Steps 1 to 8 apply. By Lemma 14.1, G has no odd cycle. By Lemma 4.9, G is bipartite. □

Proof of Corollary B. Only if. An orientation of $L(G)$ that is kernel-perfect and has every out-degree at most $k-1$ is a k-certificate *[Definition 2.5]*, so G is bipartite by Theorem A.

If. A k-regular graph has maximum degree k, so Cited Theorem 2.6 gives the orientation. □

Remark 15.1 (what the proof used). Besides elementary counting and parity, the proof of Theorem A used only three facts about kernels:

(i) kernel-perfect tournaments are linear orders (Lemma 5.8), which gives ranks;

(ii) an odd directed cycle has no kernel (Lemma 5.7), which excludes odd rings;

(iii) kernel-perfectness passes to induced subdigraphs (Lemma 5.6).

It did not use that G is of class one, it did not use Galvin's theorem or the kernel lemma, and it did not use the harder half of the Borodin, Kostochka and Woodall characterisation [4]. In fact kernel-perfectness was used only for two kinds of vertex sets: the stars $E(u)$ and their subsets (Step 1), and the edge sets of odd cycles of G (Step 3). So the proof shows more: if G is simple and k-regular and $L(G)$ has an orientation D with every out-degree at most $k-1$ such that every D_u is kernel-perfect and, for every odd cycle of G, the subdigraph of D induced by its edges has a kernel, then G is bipartite.

Corollary 15.2. *Under Standing Hypothesis 6.1: every edge has out-degree exactly $k-1$ in D, every tournament D_u is a linear order, and $r_u(v) + r_v(u) = k-1$ for every edge uv.*

Proof. Lemmas 7.1, 7.3 and 8.1. □

16 Worked examples

The numbers in this section are printed by `lecc-small/reverse_galvin_checks.py` (Section 20).

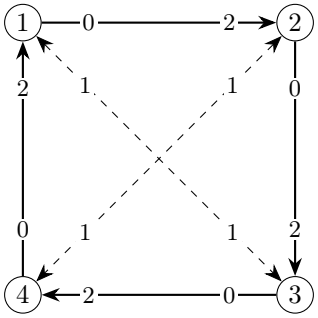
Example 16.1 (the 5-cycle, $k = 2$)**.** Let $G = C_5$ with vertices $1, 2, 3, 4, 5$ in cyclic order. Its line graph is again a 5-cycle, on the edges $12, 23, 34, 45, 51$. A 2-certificate needs out-degree at most 1 everywhere. By Step 2 every out-degree is exactly 1, so D is a directed 5-cycle in one of the two directions, and by Lemma 5.7 it has no kernel. The brute-force search agrees: 2 orientations with out-degree at most 1, none kernel-perfect. In the language of the proof: each vertex ranks one edge 0 and the other 1; tightness says each edge has rank 0 at exactly one end; so O is a directed 5-cycle, which is an odd ring. (C_5 is of class two, so here $\mathrm{ch}'(C_5) = 3 > 2$ for the trivial reason as well.)

Example 16.2 (K_4, $k = 3$)**.** K_4 is 3-regular and of class one, and it is not bipartite. By Theorem A it has no 3-certificate; the brute-force search finds 38 orientations of $L(K_4)$ with every out-degree at most 2, none kernel-perfect. Here is how the proof finds the obstruction in one of them.

Take vertices $1, 2, 3, 4$ around a square, and let every vertex rank its clockwise successor 0, the opposite vertex 1, and its predecessor 2:

w	rank 0	rank 1	rank 2
1	2	3	4
2	3	4	1
3	4	1	2
4	1	2	3

Let D orient each star by these ranks (higher rank points to lower rank, as in Lemma 7.3(b)). By Lemma 7.3(c), $d_D^+(uv) = r_u(v) + r_v(u)$, and tightness holds: for example $r_1(2) + r_2(1) = 0 + 2 = 2$ and $r_1(3) + r_3(1) = 1 + 1 = 2$. So D is an orientation of $L(K_4)$ with every out-degree equal to 2, satisfying the conclusions of Steps 1 and 2. It is not kernel-perfect, and the proof shows why.



In the figure the small number next to vertex w on the edge wv is $r_w(v)$. The arrows show the digraph O of Step 4: the middle rank is $\frac{k-1}{2} = 1$, so the sides of the square, with ranks 0 and 2 at their ends, get one arc each, and the diagonals, with rank 1 at both ends, get arcs in both directions (dashed). Every vertex has 2 out-arcs and 2 in-arcs, as Lemma 10.2(d) says.

- *Step 6 fails.* $1 \to 2 \to 3 \to 1$ is a directed cycle of O of length 3: the arcs $1 \to 2$ and $2 \to 3$ are sides, and $3 \to 1$ is a diagonal.
- *Step 5: it is a ring.* Vertex 1 prefers 2 (rank 0) to 3 (rank 1); vertex 2 prefers 3 (rank 0) to 1 (rank 2); vertex 3 prefers 1 (rank 1) to 2 (rank 2).
- *Step 3: the ring gives a kernel-less triangle in D.* At 2, the edge 12 (rank 2) points to 23 (rank 0); at 3, the edge 23 (rank 2) points to 31 (rank 1); at 1, the edge 31 (rank 1) points to 12 (rank 0). So $12 \to 23 \to 31 \to 12$ is a directed triangle in D, and $D[\{12, 23, 31\}]$ has no kernel.

Example 16.3 ($K_{3,3}$: where certificates exist)**.** $K_{3,3}$ is 3-regular and bipartite, so Galvin's theorem gives 3-certificates. The brute-force search finds 148 orientations of $L(K_{3,3})$ with every out-degree at most 2, of which exactly 12 are kernel-perfect. There are exactly 12 proper edge colourings of $K_{3,3}$ with colours $0, 1, 2$ (they are the 3×3 Latin squares), and the search confirms that the 12 certificates are exactly the 12 Galvin orientations, as Theorem 17.3 predicts. Steps 1 to 8 all hold for these certificates; Step 8 simply confirms that every cycle of $K_{3,3}$ is even.

Example 16.4 (the diamond: regularity is needed)**.** The diamond is K_4 minus one edge. It has degrees $3, 3, 2, 2$, so it is not regular; it contains a triangle, so it is not bipartite; it has $\Delta = 3$. The brute-force search finds 72 orientations of its line graph with every out-degree at most 2, and exactly 2 of them are kernel-perfect. So it has 3-certificates, and Theorem A fails without regularity. This agrees with McDonald [10, Theorem 3], since the diamond is a 4-cycle plus one edge, a bipartite graph plus an edge. In the proof, Step 2 is what breaks: the slack of Remark 8.2 is 2 (four edges join a vertex of degree 3 to one of degree 2, each contributing $\frac{1}{2}$), and indeed none of the 72 orientations has every out-degree equal to 2.

17 Rigidity: every certificate is Galvin's

Definition 17.1 (Galvin orientation)**.** Let G be a simple bipartite graph with bipartition (A, B), and c a proper edge colouring of G with colours in $\{0, \dots, k-1\}$. The *Galvin orientation* D_c of $L(G)$ is defined as follows. Let e, f be adjacent edges and $w = w(e, f)$ their common end (Lemma 4.11(b)); $c(e) \neq c(f)$ because c is proper.

- If $w \in A$: $e \to f$ iff $c(e) > c(f)$ (point to the smaller colour).
- If $w \in B$: $e \to f$ iff $c(e) < c(f)$ (point to the larger colour).

Swapping the roles of A and B gives the same family of orientations: it equals $D_{c'}$ for the colouring $c' = k - 1 - c$. So the convention does not matter.

Lemma 17.2. *If G is simple, k-regular and bipartite, then in every Galvin orientation D_c each edge has out-degree exactly $k - 1$.*

Proof. (1) Let $e = ab$ with $a \in A$, $b \in B$. At a there are k edges with k distinct colours from $\{0, \dots, k-1\}$, so every colour occurs at a exactly once. The same holds at b.

(2) At a, e points to the edges of smaller colour: there are $c(e)$ of them.

(3) At b, e points to the edges of larger colour: there are $k - 1 - c(e)$ of them.

(4) Every out-neighbour of e is at a or at b, not both *[Lemma 4.11(c)]*. So $d^+(e) = c(e) + (k - 1 - c(e)) = k - 1$. □

By Galvin's theorem, as proved in [6], every D_c is kernel-perfect, so by Lemma 17.2 every D_c is a k-certificate. The next theorem is the converse.

Theorem 17.3 (rigidity)**.** *Let $k \geq 1$ and let G be a simple k-regular bipartite graph with bipartition (A, B).*

(a) For every k-certificate D of G, the map $c(ab) = r_a(b)$ (for $a \in A$, $b \in B$) is a proper edge colouring with colours $0, \dots, k-1$, and $D = D_c$.

(b) This c is the only proper colouring with $D = D_c$.

(c) Consequently, using Galvin's theorem, $c \mapsto D_c$ is a bijection from the proper edge colourings of G with colours $0, \dots, k-1$ onto the k-certificates of G.

Proof. (a) Let D be a k-certificate. Standing Hypothesis 6.1 holds, so Steps 1 and 2 apply: ranks exist and $r_a(b) + r_b(a) = k - 1$ on every edge.

(1) *c is proper at vertices of A.* On $E(a)$, c equals r_a, which is injective *[Lemma 7.3(a)]*.

(2) *c is proper at vertices of B.* For $ab \in E(b)$, $c(ab) = r_a(b) = k - 1 - r_b(a)$ *[Lemma 8.1]*. The map r_b is injective on $E(b)$, and so is $x \mapsto k - 1 - x$. So c is injective on $E(b)$.

(3) *D agrees with D_c at vertices of A.* For distinct $e, f \in E(a)$: $e \to f$ in D iff $r_a(e) > r_a(f)$ *[Lemma 7.3(b)]* iff $c(e) > c(f)$, which is the rule of D_c at $a \in A$.

(4) *D agrees with D_c at vertices of B.* For distinct $e, f \in E(b)$: $e \to f$ in D iff $r_b(e) > r_b(f)$ iff $k - 1 - c(e) > k - 1 - c(f)$ *[(2)]* iff $c(e) < c(f)$, which is the rule of D_c at $b \in B$.

(5) Every pair of adjacent edges has exactly one common end *[Lemma 4.11(b)]*, which lies in A or in B. By (3) and (4), D and D_c have the same arc between them. So $D = D_c$.

(b)

(1) Suppose $D = D_{c'}$ for a proper colouring c'. Let $a \in A$ and $e \in E(a)$.

(2) In $D_{c'}$, the edge e points, inside $E(a)$, to exactly the edges of smaller colour. The colours on $E(a)$ are $0, \ldots, k-1$, each once (as in step (1) of the proof of Lemma 17.2), so there are $c'(e)$ such edges.

(3) That number is $r_a(e)$ by Definition 7.2. So $c'(e) = r_a(e) = c(e)$.

(4) Every edge has an end in A, so $c' = c$.

(c) By Galvin's theorem and Lemma 17.2, $c \mapsto D_c$ maps proper colourings to k-certificates. It is injective by (b) (if $D_c = D_{c'}$ then $c = c'$) and onto by (a). □

Remark 17.4. Parts (a) and (b) do not use Galvin's theorem: they say that if a certificate exists, it has Galvin's form. Only (c), the statement that every D_c really is kernel-perfect, uses it. The brute-force search confirms the bijection of (c) on C_4, C_6, $K_{3,3}$ and the cube Q_3, with 2, 2, 12 and 24 certificates and as many colourings.

18 Orientations with arcs in both directions

Some authors, for example McDonald and Puleo [11], apply the kernel lemma to digraphs in which adjacent vertices may be joined by arcs in both directions. The kernel lemma holds in that setting too: the proof in [1, Chapter 33] never uses that two vertices are joined by at most one arc. Theorem A extends to this setting.

Proposition 18.1. *Let $k \geq 1$ and let G be a simple k-regular graph. Let D be a digraph with vertex set E such that*

(i) every arc of D joins two edges that are adjacent in $L(G)$;

(ii) every two edges adjacent in $L(G)$ are joined by at least one arc (possibly by both);

(iii) D is kernel-perfect;

(iv) $d^+_D(e) \leq k-1$ for every e.

Then no two edges are joined by arcs in both directions. So D is a k-certificate, and G is bipartite.

Proof. (1) Let p be the number of adjacent pairs joined by both arcs. By (i) and (ii), the number of arcs of D is (number of edges of $L(G)$) $+ p$.

(2) By Lemma 5.3 and (iv), the number of arcs is $\sum_e d^+_D(e) \leq |E|(k-1)$.

(3) The number of edges of $L(G)$ is $|E|(k-1)$ *[Lemma 4.11(d)]*.

(4) So $|E|(k-1) + p \leq |E|(k-1)$, hence $p = 0$.

(5) Then D is an orientation of $L(G)$ satisfying (C1) and (C2), and Theorem A applies. □

19 Impact and consequences

This section explains what the result implies and what it leads to. Statements about other papers are cited, not proved.

19.1 What it adds to Galvin's theorem

Galvin's theorem says: bipartite ⇒ the kernel method with lists of size Δ works. Theorem A says: for simple regular graphs, the method works ⇒ bipartite. Theorem 17.3 says: when it works, the only certificates are Galvin's own. Together they describe the kernel method with lists of size Δ on simple regular graphs completely:

- it succeeds exactly on the bipartite graphs (Corollary B);
- where it succeeds, the certificates are exactly the Galvin orientations, one for each proper Δ-edge colouring (Theorem 17.3).

So, on simple regular graphs, Galvin's 1995 construction is not one successful use of the kernel method among possibly many: it is all of it. Section 2.6 explains why this was not a foregone conclusion.

The size of the contribution should be stated plainly. It is a *barrier* result, about the limits of a method. It proves the LECC for no new graph, and it has an elementary proof of about a page.

19.2 Which cases of the conjecture it affects

For a simple regular graph G there are three cases.

(1) *G bipartite.* The LECC holds by Galvin's theorem, and the kernel method is the proof.

(2) *G of class two.* Then $\mathrm{ch}'(G) \geq \chi'(G) = \Delta + 1$, so the LECC asks for $\mathrm{ch}'(G) = \Delta + 1$, a question about lists of size $\Delta + 1$. Theorem A says nothing here.

(3) *G non-bipartite and of class one.* The LECC predicts $\mathrm{ch}'(G) = \Delta$. Theorem A says the kernel method cannot prove this, for any such G.

Case (3) is a large part of the conjecture. It contains:

- *Complete graphs of even order.* K_{2n} ($n \geq 2$) is $(2n-1)$-regular, of class one, and not bipartite. The LECC for K_{2n} is open in general; it is known when $2n-1$ is an odd prime (Schauz [12]) and for further families (Jafari [7]), all by algebraic methods. Theorem A explains why no kernel proof has appeared: none exists. It contains our earlier result that $L(K_{2n})$ has no $(2n-1)$-certificate (`list-colouring/12-theorem-even-complete.md`), and strengthens, for complete graphs of even order, McDonald's result [10], which excludes only certificates of Galvin's form. (K_{2n+1} is of class two, case (2).)
- *Planar regular class-one graphs that are not bipartite*, such as K_4, the octahedron, the triangular prism and the pentagonal prism $C_5 \square K_2$. Ellingham and Goddyn [5] proved the LECC for every d-regular d-edge-colourable planar graph, by the Alon and Tarsi method [2]. By Theorem A, the kernel method could not have proved it for the non-bipartite ones. The brute-force search confirms this directly for K_4, the octahedron and both prisms (Section 20).

- *Schauz's open small graphs.* Schauz [13] proved the LECC for most graphs on at most 10 vertices by the Alon and Tarsi method. He left open the regular class-one graphs on which the relevant coefficient vanishes: 3 on 8 vertices and 51 on 10. We reproduced this census (`lecc-small/RESULTS.md`, R1). Of these, 52 are covered by no known theorem, and none of the 52 is bipartite, so by Theorem A the kernel method fails on every one of them.
- *Cubic class-one graphs with a triangle.* Our earlier triangle theorem (`list-colouring/11-triangle-theorem.md`) says these have no 3-certificate. It is the special case of Theorem A with $k = 3$ and an odd cycle of length 3.

19.3 Both standard methods fail on the same open cases

The two standard tools for the LECC are the kernel method and the Alon and Tarsi polynomial method [2]. For a k-regular graph G, the Alon and Tarsi method with lists of size k also has very little room, for a counting reason of the same kind as Step 2. The graph polynomial of $L(G)$ is a product of one linear factor per edge of $L(G)$, so it is homogeneous of degree $|E|(k-1)$ (Lemma 4.11(d)) in $|E|$ variables. A monomial usable with lists of size k has every exponent at most $k-1$, and its total degree is $|E|(k-1)$, so every exponent must equal $k-1$. Only one monomial is usable, and on Schauz's open graphs its coefficient is zero. So:

- on the 52 open graphs, *both* standard methods provably fail: Alon and Tarsi by the census, the kernel method by Theorem A. This explains why they are open, and it is why a third method was needed. A SAT-based search over list assignments (`lecc-small/RESULTS.md`; the general approach is that of [8]) has since settled 6 of them;
- the two methods are *incomparable* on regular graphs. Alon and Tarsi proves $\mathrm{ch}'(K_4) = 3$ (a case of the Ellingham and Goddyn theorem), where the kernel method cannot. In the other direction, two of Schauz's zero-sum graphs on 10 vertices are bipartite, one of them $K_{5,5}$ (`lecc-small/RESULTS.md`, R2): there the uniform coefficient is zero, so Alon and Tarsi with lists of size k fails, while Galvin's kernel method works.

Any attack on case (3) must therefore start from the Alon and Tarsi side in a stronger form, or from a new idea.

19.4 The limits of the statement

- *Not a lower bound on* ch'. It does not say $\mathrm{ch}'(G) > \Delta$; K_4 shows that would be false (Section 2.5).
- *Regular graphs only.* The diamond shows this is necessary (Example 16.4); regularity is used exactly once, in Step 2 (Remark 8.2).
- *Lists of size* Δ *only.* With lists of size $\Delta + 1$ the kernel method succeeds on non-bipartite regular graphs: the brute-force search finds 6 kernel-perfect orientations of $L(K_4)$ and 648 of the line graph of the triangular prism with every out-degree at most 3 (Section 20). McDonald [10, Theorem 4] shows the Petersen graph also has one. For class-one graphs this gives only $\mathrm{ch}' \leq \Delta + 1$, not the conjectured value.
- *Simple graphs only.* The multigraph case is open (Remark 7.4).
- *The kernel lemma only.* It says nothing about other methods that use orientations, such as Alon and Tarsi.

19.5 Questions it leads to

Question 19.1 (multigraphs)**.** Is Theorem A true for regular multigraphs? Galvin's theorem holds for bipartite multigraphs, so the natural statement is again a characterisation. The proof breaks at Lemma 7.3(c), where two parallel edges share two stars.

Question 19.2 (the least budget)**.** For a non-bipartite regular class-one graph G, what is the least t such that G has a t-certificate? Theorem A says $t \geq \Delta + 1$, and for K_4 and the triangular prism the search shows $t = \Delta + 1$. Is $t = \Delta + 1$ for every such graph? This is the regular-graph version of McDonald's question [10] for Galvin orientations, asked for all orientations.

Question 19.3 (near-regular graphs)**.** Remark 8.2 measures the distance from regularity by the slack $\sum_{uv}\big(\Delta - \frac{d(u)+d(v)}{2}\big)$. When the slack is small, almost every out-degree is forced to be $\Delta - 1$. Is there a quantitative version of Theorem A, for example one in which an odd cycle far from the vertices of degree less than Δ forbids a Δ-certificate?

Question 19.4 (a method for case (3))**.** Both standard methods fail on Schauz's 52 open graphs. Is there a general method that works on non-bipartite regular class-one graphs, or a structural reason that makes them hard?

20 Computational confirmation

The proof does not depend on any computation. The computations below check it independently.

Brute force, using no lemma. The script `lecc-small/reverse_galvin_checks.py` (pure Python, about 30 seconds) does the following for each graph G in Table 1, with $k = \Delta(G)$:

(1) it lists every orientation of $L(G)$ with every out-degree at most $k - 1$;

(2) it tests each one for kernel-perfectness directly from Definition 2.3, by searching every induced subdigraph for a kernel;

(3) it counts the proper edge colourings with colours $0, \ldots, k - 1$;

(4) for the regular bipartite graphs, it checks that the set of certificates equals the set of Galvin orientations D_c and that $c \mapsto D_c$ is injective (Theorem 17.3).

It uses none of the lemmas of this document. Its output is `lecc-small/data/reverse_galvin_checks.out`.

What the table shows:

- *Theorem A:* every regular non-bipartite graph has 0 certificates, including the class-one graphs K_4, both prisms and the octahedron.
- *Corollary B and Theorem 17.3:* every regular bipartite graph has certificates, exactly as many as colourings, and they are the Galvin orientations (the script reports "ok" for each).
- *Step 2:* for every regular graph the two orientation counts agree: every orientation with out-degrees at most $k - 1$ has all out-degrees exactly $k - 1$. For the non-regular diamond they do not.
- *Regularity is needed:* the diamond has 2 certificates.

The same script also counts kernel-perfect orientations with every out-degree at most k (lists of size $k + 1$): 6 for K_4 (out of 2608 orientations) and 648 for the triangular prism (out of 132864).

G	k	regular	bipartite	class	orient.	all $d^+ = k-1$	certificates	colourings
C_4	2	yes	yes	1	2	2	2	2
C_5	2	yes	no	2	2	2	0	0
C_6	2	yes	yes	1	2	2	2	2
K_4	3	yes	no	1	38	38	0	6
$K_{3,3}$	3	yes	yes	1	148	148	12	12
prism $K_3 \square K_2$	3	yes	no	1	154	154	0	6
cube Q_3	3	yes	yes	1	600	600	24	24
$C_5 \square K_2$	3	yes	no	1	2466	2466	0	30
Petersen	3	yes	no	2	2336	2336	0	0
K_5	4	yes	no	2	47504	47504	0	0
octahedron	4	yes	no	1	325744	325744	0	48
diamond	3	no	no	1	72	0	2	6

Table 1: Brute force. "orient." counts orientations of $L(G)$ with every out-degree at most $k-1$; "all $d^+ = k-1$" counts those in which every out-degree equals $k-1$; "certificates" counts the kernel-perfect ones; "colourings" counts proper edge colourings with colours $0, \ldots, k-1$ (so it is 0 exactly for class two). The class column follows from the colourings column.

Larger census, using the odd-ring reduction. The SAT search `lecc-small/kernel_sat.py` decides whether a k-certificate exists for every connected regular graph on 6 and 8 vertices (every degree from 3), on 10 vertices (degrees 3 to 7), for cubic graphs on 12 vertices, triangle-free cubic graphs on 14 vertices and 4-regular graphs on 12 vertices: 1923 graphs. A certificate exists exactly for the 29 bipartite ones (`lecc-small/data/kernel_all_small.csv`, `lecc-small/data/kernel_N12_k4.csv`). This search encodes Steps 1 and 2 and the full odd-cycle characterisation of Borodin, Kostochka and Woodall [4], so it is not independent of the ideas of the proof, but it does not use Steps 4 to 8.

21 Novelty and status

We have not found Theorem A or Theorem 17.3 in the literature. The search is recorded in `lecc-small/05-novelty-check.md`. We checked:

- Borodin, Kostochka and Woodall [4]: they characterise which orientations of a line graph are kernel-perfect, not which graphs admit one within the budget Δ;
- McDonald [10]: negative results only for Galvin's special form of orientation, for K_n and for one non-regular graph;
- McDonald and Puleo [11]: consistent with Proposition 18.1; no non-bipartite regular graph of degree at least 3 lies in their class;
- the papers citing [4], and recent work on the LECC for complete graphs [7].

Two sources could not be checked: the 2016 Auburn M.Sc. thesis of A. Blumenthal, *On Galvin's Theorem and Stable Matchings* [3], and Woodall's 2001 survey [15]. Since the proof is short and elementary, the result may be known to specialists without being published. It should be described as "not found in the literature", not as new, until those two sources are checked.

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